2.3 Solving Trigonometric Equations

Name _____ Period ____ Date _____

Pre-Calculus 2.3 (30-minute plan) · Student sheet

I can…

- solve a trig equation and give every solution, either the general solution or the solutions in [0, 2π).

Part 1 · Notes: fill in the blanks

1. Isolate the _____. 2. Extracting square roots: remember the _____.

3. Factor. Never _____ both sides by a trig function: you lose solutions.

4. Quadratic type: factor as if sin x were a _____. 5. If you square both sides, _____ every answer.

General solution: add _____ for sine and cosine, and add _____ for tangent.

Part 2 · Examples (work them with the class)

Example 1

Solve: $2 \sin x - 1 = 0$

Example 2

Solve: $3 \tan^2 x - 1 = 0$

Example 3

Solve on [0, 2π): $2 \sin^2 x - \sin x - 1 = 0$

Part 3 · Partner practice

Partner practice

Solve: $\sqrt{3} \csc x - 2 = 0$

Part 4 · On your own: choose 2 of 3

Problem 1

Solve on $[0, 2\pi)$: $2\cos x + 1 = 0$

Problem 2

Solve on $[0, 2\pi)$: $\tan^2 x - 3 = 0$

Problem 3

Solve on $[0, 2\pi)$: $2\cos^2 x + \cos x - 1 = 0$